

Q-Orbit — Phase 1, Deliverable 4: Proof-Profile Comparison and Recommendation

Document ID: QO-D4-001 | **Version:** 1.0 | **Date:** 2026-08-27 **Status:** THEORETICAL / NOT PHYSICALLY VALIDATED — PRIVATE-BLOCKED **Evidence labels:** NUMERICALLY-VERIFIED (locked fixture facts, Canonical Facts Record) ·EV-5 LITERATURE-SUPPORTED (verified citation record; never device evidence) ·ASSUMPTION-DEPENDENT. All citations per the D3 verification ledger (QO-D3-001), including the post-audit upgrade: Currás-Lorenzo et al. unified framework = **Optica Quantum 3, 525 (2025)**; the Tupkary et al. consolidated decoy-BB84 proof = **arXiv:2601.18035 (2026, preprint — under review)**; Tan & Nahar = **PRX Quantum 7, 020342 (2026), DOI 10.1103/f42p-524t** (genuine new-format APS DOI). (Red-team C1: an earlier claim that arXiv:2601.18035 was published as Quantum 10, 2037 (2026) was checked and removed as false/unverified; it is cited as a preprint throughout.)

1. Purpose

This document compares candidate security proofs/frameworks for Q-Orbit’s protocol class — efficient-BB84, weak-coherent pulses, 1 signal + 2 decoy intensities (one vacuum), satellite downlink, finite key — against the frozen V0.16 fixture and the package’s unmapped-effects register, and issues the Phase 1 recommendation required by the D4 specification. The frozen reference point (NUMERICALLY-VERIFIED): margin $M = s_{X,0} + s_{X,1} \cdot [1 - h_2(_X)] - _EC - 6 \cdot \log_2(21/_s) - \log_2(2/_c)$; baseline $M = 41,338.62418456675$ bits at 102 s half-window; $n_X = 492,818.0901525894$ per-pass block; deterministic screen 568 positive / 1,113 non-positive of 1,681 grid points. These are model-computation facts; no physical claim follows from them.

2. Comparison matrix

Column definitions: **Protocol compatibility** = fit to efficient-BB84 WCP 3-intensity satellite downlink without protocol change. **Finite-key?** = composable bound at finite block length. **Imperfect phase randomization** / **Source flaws** / **Correlations** / **Detector mismatch** = native coverage of that imperfection class (“native” = inside the proof; “bolt-on” = via an external parameter; “no” = assumption required). **Required characterization** = the measured inputs the proof needs before any number is a security statement. **Integration difficulty** = cost of adoption relative to the frozen V0.16 fixture. **Suitability for Q-Orbit** = verdict under the package’s fail-closed, zero-characterization boundary.

Proof / frame- work	Protocol compati- bility	Finite- key?	Imperfect phase random- ization	Source flaws	Correlations	Detector mis- match	Required charac- teriza- tion	Main assump- tions	Integratio difficulty
Frozen Sidhu/ Lim V0.16 profile (refer- ence row) — Sidhu et al., npj QI 8, 18 (2022); Lim et al., PRA 89, 022307 (2014)	Native — exactly the fixture’ s protocol and algebra	Yes (union bound over 21 events; penalty 256.5669430839006 bits, NUMERICALLY- VERIFIED)	No (assumed perfect)	No (GLLP Δ term absent in fixture)	No (IID assumed)	No (scalar , squash- ing assumed)	Point- valued {_j}, , p_ec, p_ap, intrinsic QBER, loss curve — all currently AS- SUMED	Perfect PR; IID pulses; exact in- tensities; qubit/ squash- ing detec- tion; no side channels; point pa- rameters	Zero —is the fixture
Lim et al. 2014 stan- dalone —PRA 89, 022307	Native (3- intensity decoy, efficient BB84)	Yes (tight, concise bounds)	No	No	No	No	Same point- valued inputs	Same as reference row	Zero (already instanti- ated)

Table 1 – continued

Proof / frame- work	Protocol compati- bility	Finite- key?	Imperfect phase random- ization	Source flaws	Correlations	Detector mis- match	Required charac- teriza- tion	Main assump- tions	Integrati- on difficulty
Lo-Ma- Chen 2005 + Ma et al. 2005 –PRL 94, 230504; PRA 72, 012326	Native conceptu- ally (decoy invention + practical fluctua- tion treat- ment)	Partial (statistical- fluctuation treat- ment; not a compos- able finite-key bound)	No	No	No	No	Known intensi- ties; PR source	PR WCP; IID; ideal qubits	Low (already the decoy layer of the fixture)
GLLP 2004 – Quan- tum Inf. Comput. 4, 325	Protocol- agnostic (P&M with imperfect devices)	No (asymp- totic)	No	Bolt-on only: basis- independent flaws via single Δ	No	Partial (basis- independent detection assumed; mis- match not native)	Measured Δ (basis- independent flaw bound)	Basis- independent flaws; squash- ing	Low (single additive term)

Table 1 – continued

Proof / frame- work	Protocol compati- bility	Finite- key?	Imperfect phase random- ization	Source flaws	Correlations	Detector mis- match	Required charac- teriza- tion	Main assump- tions	Integratio difficulty
Loss- tolerant Tamaki 2014 + Mizu- tani 2015 finite- key – PRA 90, 052314; NJP 17, 093011	Requires 3-state encoding modifica- tion (Tamaki); Mizutani extends finite-key + fluctu- ating intensi- ties	Yes (Mizu- tani)	No	Native (unchar- acterized state- preparation flaws without basis- independence)	No (IID); intensity fluctua- tion intervals native (Mizu- tani) (Mizu- tani) (independence)	No	3-state overlaps / state charac- teriza- tion; intensity intervals [,]	Characterized reference states; IID otherwise	Medium (modified encoding + new phase- error estima- tion)
Nahar- Upad- hyaya- Lütken- haus 2023 (seed #2) – PR Applied 20, 064031	Native (general- ized decoy- state BB84)	Yes (finite- key- relevant decoy analysis)	Native (charac- terized phase PDF enters decoy bounds)	No (PR only)	No	No	Measured global- phase distribu- tion inside the ad- missible class	IID; known in- tensities; detection ideal	Medium

Table 1 – continued

Proof / frame- work	Protocol compati- bility	Finite- key?	Imperfect phase random- ization	Source flaws	Correlations Detector	Required charac- teriza- tion	Main assump- tions	Integrati- on difficulty
Quantum- coin / finite- key imperfect- phase line — Pereira et al., PRR 5, 023065 (2023) (modified BB84); Currás- Lorenzo et al., QST 9, 015025 (2023); unified frame- work Currás- Lorenzo et al., Optica Quan- tum 3, 525 (2025)	Native to BB84- family; unified frame- work handles flaws jointly	Yes (finite- key analyses exist in the line)	Native	Native (SPFs + side channels jointly in the unified frame- work)	No (within- line); pair with correlation- tolerant row	No (source- side focus) PDF; side- channel distin- guishabil- ity budget	IID pulses; detection model	Medium- high (more moving parts; per-flaw theorems unless unified frame- work used)

Table 1 – continued

Proof / frame- work	Protocol compati- bility	Finite- key?	Imperfect phase random- ization	Source flaws	Correlations	Detector mis- match	Required charac- teriza- tion	Main assump- tions	Integrati- on difficulty
Correlation- tolerant decoy – Zapatero et al., Quantum 5, 602 (2021); Sixto et al., PR Applied 18, 044069 (2022); un- bounded: Pereira et al., QST 10, 015001 (2025); experi- mental reality: Yoshino et al., npj QI 4, 8 (2018)	Native (decoy- state BB84 with cor- related intens- ties)	Yes	No	No	Native (bounded nearest- neighbour → corre- lated fluctua- tions → un- bounded)	No	Correlation length and correlation- magnitude bounds (mea- sured condi- tional intens- ties)	Bounded/ charac- terized correla- tion; otherwise standard	Medium

Table 1 – continued

Proof / frame- work	Protocol compati- bility	Finite- key?	Imperfect phase random- ization	Source flaws	Correlations	Detector mis- match	Required charac- teriza- tion	Main assump- tions	Integrati- on difficulty
Consolidated decoy- BB84 with imper- fections — Tupkary et al., arXiv:2601.18035 (2026, preprint —under review; not peer- reviewed)	Alternative (rigorous consoli- dated decoy- BB84 proof)	Yes	Yes (via incorporated imperfec- tion terms)	Yes (con- solidated source- imperfection treat- ment)	Partial (per the consoli- dated scope; pair with correlation- tolerant row for long- range memory)	Partial	Same imperfec- tion parame- ters as the F2/ F5 lines, measured	Standard detection model with bounded devia- tions	Medium (estima- tion LP shared with Lim family struc- ture)
MDI- QKD — Lo, Curty, Qi, PRL 108, 130503 (2012); Braun- stein & Piran- dola, PRL 108, 130502 (2012)	Incompatible with the direct satellite down- link (see verdict below)	Yes (finite- key MDI: Curty et al., Nat. Commun. 5, 3732 (2014))	Addressable in MDI variants	Addressable	Addressable in variants	Eliminates detector trust archi- tect- urally	Source- side only	Untrusted relay BSM; charac- terized sources	Very high (architec- ture change)

Table 1 – continued

Proof / frame- work	Protocol compati- bility	Finite- key?	Imperfect phase random- ization	Source flaws	Correlations Detector	Required mis- match	Required charac- teriza- tion	Main assump- tions	Integratio difficulty
Numerical SDP – Winick, Lütken- haus, Coles, Quan- tum 2, 77 (2018); George, Lin, Lütken- haus, PRR 3, 013274 (2021); dimen- sion reduction Upad- hyaya et al., PRX Quan- tum 2, 020325 (2021); variable- length Tupkary, Tan, Lütken- haus, PRR 6, 023002 (2024); detector imperfec- tions Tupkary et al., Quan- tum 9, 1937 (2025); Open- QKDse-	Native (arbi- trary P&M proto- cols)	Yes (acceptance- test + min- tradeoff / EUR)	Native (as con- straint)	Native (arbi- trary charac- terized imperfec- tions as SDP con- straints)	Via EAT combina- tion (Kamin et al., arXiv:2406.10198, preprint)	Native (imperfect- detector numer- ics)	The most demand- ing: verified charac- teriza- tion of every imperfec- tion entering the con- straints	Squashing/ dimension- reduction validity; verified numerics (interval arith- metic)	High (tooling, verifica- tion, reduced line-by- line auditabil- ity)

Table 1 – continued

Proof / frame- work	Protocol compati- bility	Finite- key?	Imperfect phase random- ization	Source flaws	CorrelationsDetector	Required mis- match	Required charac- teriza- tion	Main assump- tions	Integrati difficulty
EAT — Dupuis, Fawzi, Renner, Commun. Math. Phys. 379, 867 (2020); Metger & Renner, Nat. Commun. 14, 5272 (2023); decoy- state via EAT: Kamin et al., arXiv:2406.10198 (2024, preprint); charac- terized devices: George et al., arXiv:2203.06554 (preprint)	Applicable to P&M decoy (recent)	Yes	Via con- straints	Via con- straints	Native (drops IID — strongest handle on memory)	Via con- straints	Min- tradeoff function inputs; charac- teriza- tion per con- straint	Markov/ round structure of the EAT theorem	High

Table 1 – continued

Proof / frame- work	Protocol compati- bility	Finite- key?	Imperfect phase random- ization	Source flaws	Correlations	Detector mis- match	Required charac- teriza- tion	Main assump- tions	Integrati- on difficulty
Postselection / de Finetti –Chris- tandl, König, Renner, PRL 102, 020504 (2009); optical- adapted: Nahar et al., PRX Quan- tum 5, 040315 (2024)	Generic lift over BB84- family	Yes (via reduc- tion)	Inherits base proof	Inherits base proof	Partial (permuta- tion symme- try require- ments)	Inherits base proof	Base proof’s inputs	Permutation symme- try of the protocol	Medium

Table 1 – continued

Proof / frame- work	Protocol compati- bility	Finite- key?	Imperfect phase random- ization	Source flaws	Correlations	Detector mis- match	Required charac- teriza- tion	Main assump- tions	Integrati- on difficulty
Composable frame- works (cross- cutting layer) — Müller- Quade & Renner, NJP 11, 085006 (2009); Port- mann & Renner, Rev. Mod. Phys. 94, 025008 (2022)	Not a proof — the definition layer all rows are stated against	Definitional	N/A	N/A	N/A	N/A	N/A	Trace- distance security defini- tion; composi- tion theorems	Zero (already the 'package' s claim lan- guage)

MDI-QKD satellite-downlink verdict (explicit, per D4 mandate). MDI-QKD removes all detector-side trust by relocating the measurement to an untrusted relay that receives light from *both* parties and performs a Bell-state measurement. In the Q-Orbit concept the satellite is the sender and the ground station is the receiver: there is no relay between two senders, and the detector side is precisely the ground station whose trust is at issue. MDI therefore applies only under architecture change: (a) *uplink MDI* — ground transmits, satellite performs the untrusted BSM — inverts the link, adds uplink loss/turbulence asymmetry, and moves optics to orbit (a different mission concept); (b) *satellite-relay MDI between two ground stations* — the satellite becomes the untrusted middle node, roughly doubling channel loss through two downlinks. **Within the direct-downlink concept, detector trust cannot be architecturally eliminated; the only available path is a trusted-but-characterized-and-bounded receiver.** MDI is logged as an architectural alternative and excluded from the Q-Orbit proof path.

3. Decision analysis

3.1 Coverage honesty statement (required)

No single existing proof covers all 15 mandated effects (of which none is MAPPED-IN-CURRENT-FIXTURE). The gap is characterization, not proof identification: for **7 of the 8 Agent-C source-side effect classes** (S1–S8 taxonomy), candidate proof treatments exist in verified literature (loss-tolerant line, generalized decoy, correlation-tolerant decoy, leaky-source line, unified framework Optica Quantum 3, 525 (2025), consolidated proof arXiv:2601.18035 (2026, preprint)); detector-side mismatch is likewise covered by mature mismatch-bounded proofs pending measured bounds. The genuine open proof problems are **detector-side correlated afterpulsing/memory** and **rate-dependent yields** (which break the decoy method’s intensity-independent-yield identity): no verified turnkey treatment exists for either in this fixture class — a literature gap, not a search failure. Every option below is therefore evaluated as *proof identification + characterization prerequisites*, never as a turnkey fix.

3.2 The three options

Option A — Retain the frozen F4/F3 profile and constrain the device. Keep the Lim 2014 / Sidhu 2022 construction as the shipping proof; harden it with (i) an explicit assumption ledger attached to every rate claim (the review arXiv:2502.10340 documents exactly which gaps reviewers hunt) and (ii) a statistics upgrade from plain Chernoff/Hoeffding to multiplicative Chernoff (Zhang et al., PRA 95, 012333 (2017)), Kato’s inequality (arXiv:2002.04357, preprint — as used in the CubeSat-scale analysis Islam et al., PRX Quantum 5, 030101 (2024)), or the sharp finite statistics of Mannalath, Zapatero, Curty, PRL 135, 020803 (2025). *Assumption coverage*: unchanged — all six assumption classes of D3 §3 remain assumed; imperfection coverage limited to a GLLP Δ bolt-on at best. *Integration cost*: lowest; the locked numerics instantiate this family natively (zero rewrite of the margin structure). *Fail-closed compatibility*: highest — closed-form, line-by-line auditable, and the assumption ledger makes every conditional explicit rather than hidden.

Option B — Adopt a generalized (imperfection-native) proof. Re-base the security claim on the complementarity/phase-error family with native imperfections: loss-tolerant state-preparation coverage (Tamaki PRA 90, 052314 (2014); Mizutani NJP 17, 093011 (2015)), generalized decoy for imperfect phase randomization (seed #2; Currás-Lorenzo QST 9, 015025 (2023)), unified source-imperfection framework (Optica Quantum 3, 525 (2025)), anchored on the consolidated rigorous decoy-BB84 proof (Tupkary et al., arXiv:2601.18035, 2026 preprint). *Assumption coverage*: converts Q-Orbit’s source-imperfection citations into covered proof terms — the single largest coverage gain available without architecture change. *Integration cost*: medium (months-scale); the estimation linear program is shared with the Lim-family structure, so the margin model survives structurally. *Fail-closed compatibility*: good, provided every new proof term is fed by a characterization interval with a stated ϵ_j and the software refuses numeric evaluation of uncharacterized terms (Category-2/3 guards).

Option C — Layered architecture. Separate concerns: a satellite finite-key analytic proof at the optical/count layer; implementation-security proof extensions supplying admissible parameter bounds; and a characterization/certification composition layer per Tan & Nahar (PRX Quantum 7, 020342 (2026)) turning point assumptions into interval-conditioned, epsilon-composed claims. *Assumption coverage*: the only option that addresses A6 (point-valued parameters) and the missing ϵ_{char} / ϵ_{auth} budget classes structurally. *Integration cost*: highest in total, but decomposable into stages. *Fail-closed compatibility*: by construction — each layer emits labeled, conditional outputs, and the composition layer is exactly the certify-then-run joint bound $\Pr[\text{certification approves AND key insecure}] = \epsilon_{\text{char}} + \epsilon_{\text{protocol}}$.

3.3 Synthesis

The options are complementary, not exclusive: A is the only zero-rewrite, satellite-standard, red-team-auditable shipping proof available in Phase 1; B is the minimum-change path to honest source-imperfection coverage; C is the architecture in which A and B become security *statements* rather than conditional numbers. Detector-side correlated memory and rate-dependent yields remain open proof problems under every option and must be carried as explicit limitations.

4. RECOMMENDATION

Adopted (per the Agent B recommendation, confirmed here):

1. **Profile A, hardened, is the V0.17-TA1 shipping proof.** The frozen F4/F3 construction (Lim 2014 / Sidhu 2022 lineage) is retained as the shipping analytic proof because it is the family the locked, NUMERICALLY-VERIFIED numerics already instantiate, it is the satellite standard (Sidhu et al. 2022; Islam et al. 2024; Micus anchor Liao et al., Nature 549, 43 (2017)), and it is the most auditable under the Phase 1 red-team gate. Two mandatory hardening actions: (i) an **assumption ledger per claim** — every rate statement carries its six assumption classes (D3 §3) explicitly; (ii) a **statistics upgrade** to Kato's inequality or Mannalath–Zapatero–Curty sharp statistics (drop-in at the fixture level), the cheapest way to shrink the 256.57-bit-class penalty without changing proof family. All Profile A outputs remain CONDITIONAL-COMPUTATION-labeled while characterization is absent.
2. **Profile B is the V0.18+ upgrade path.** The consolidated decoy-BB84 proof with imperfections (Tupkary et al., arXiv:2601.18035 (2026, preprint)) is the designated anchor, with the unified source-imperfection framework (Optica Quantum 3, 525 (2025)) for joint flaw coverage, converting the source-imperfection citations into covered proof terms. Adoption is gated on the characterization inputs each term requires (state overlaps, phase-PDF bound, intensity intervals, isolation/distinguishability budgets) — until then its imperfection terms remain symbolic.
3. **Numerical SDP is the independent cross-check, not the shipping proof.** OpenQKDsecurity-class numerics (Winick 2018; George 2021; Upadhyaya 2021; Tupkary PRR 6, 023002 (2024); detector-imperfection numerics Tupkary et al., Quantum 9, 1937 (2025)) are to be run as an independent cross-check of the Profile A margins at a selected set of grid points, and positioned as the certification end-state aligned with Tan & Nahar 2026. Rationale for not shipping it in Phase 1: high tooling and interval-arithmetic verification cost, reduced line-by-line auditability, and finite-key constants at $n = 5 \times 10^5$ per pass not yet superior to the analytic bound.
4. **MDI-QKD is excluded architecturally** (verdict in §2): incompatible with the direct satellite-to-ground downlink; logged as an architectural alternative only.
5. **Target layered architecture** (the Option C structure, populated by A now and B at V0.18+):

Layer	Content	Status
Layer 0 –Frozen V0.16 fixture	Immutable regression reference: locked margin equation, 12-test regression suite, 20-check audit, deterministic 1,681-point screen. Never silently mutated; V0.17 extends, it does not modify.	In place; NUMERICALLY-VERIFIED
Layer 1 –Hardened analytic finite-key (optical/count layer)	Profile A hardened: Lim/Sidhu skeleton + Kato/Mannalath statistics + per-claim assumption ledger + 21-split validator + epsilon ledger with symbolic <code>_char/ _auth</code>	V0.17-TA1 shipping proof
Layer 2 – Implementation-security proof extensions	Profile B terms (imperfect PR, SPFs, correlations, leakage) supplying admissible parameter bounds; each term symbolic until its characterization interval exists; Category-2/3 software refusals enforced	V0.18+ upgrade path
Layer 3 –Characterization/certification composition	Tan & Nahar (PRX Quantum 7, 020342 (2026)) certify-then-run composition: robust set <code>S_robust</code> , per-parameter confidence intervals, union-bound <code>_char</code> , joint failure-bound claim language only	Framework specified (Agent E bridge); execution BLOCKED pending characterization – prohibited from emitting any certified-security claim

6. **Explicit non-claims (binding on all outputs of this recommendation):** no mission success probability, availability, Tabuk performance, implementation security, certified security, procurement tolerance, hardware readiness, deployability, field readiness, or released-key claim is made or implied by adopting any proof profile. Proof adoption changes what the *theory* covers; it changes nothing about the package’s zero-characterization boundary. The two genuine open proof problems (detector-side correlated afterpulsing/memory; rate-dependent yields) are carried forward as explicit limitations into D6 and the D10 backlog.

5. Rejected alternatives, with reasons

Rejected alternative	Reason
Numerical SDP (Profile C) as the Phase 1 shipping proof	High tooling/interval-arithmetic verification cost; reduced line-by-line auditability against the red-team gate; finite-key constants at Q-Orbit's per-pass block size ($n_X = 4.93 \times 10^5$) not yet superior to the analytic bound. Retained as cross-check and certification end-state instead.
MDI-QKD as the detector-side answer	Architecturally incompatible with a direct satellite→ground downlink (no relay between two senders; the ground receiver is precisely the party whose trust is at issue). Would require an uplink or dual-downlink relay redesign —a different mission concept, not a proof choice.
GLLP Δ bolt-on as sufficient source-flaw coverage	Covers only basis- <i>independent</i> flaws; real modulator/interferometer flaws are generically basis-dependent, for which the GLLP bound is invalid. An unmeasured Δ is a hidden assumption, not coverage —precisely the hidden-substitution move the package rules prohibit.
EAT as the primary finite-key proof	Constants historically worse than the F4 union-bound at satellite block sizes; prepare-and-measure decoy instantiation still maturing (Kamin et al., arXiv:2406.10198 — preprint). Retained as the non-IID robustness route and citation, not the shipping proof.
Postselection/de Finetti as the primary coherent-attack lift	Polynomial-cost lift is typically looser than direct concentration statistics at Q-Orbit block sizes. Retained as the supporting citation for coherent-attack lifting (Nahar et al., PRX Quantum 5, 040315 (2024)).
Absorbing imperfections into existing scalars (QBER, η, p_{ext})	Prohibited hidden substitution: phase-randomization defects → QBER; source leakage → loss; detector mismatch → scalar efficiency; correlations → IID extraneous-count term. Scalar stress tests remain legitimate engineering sensitivity probes but emit no security content.
Representing nominal/datasheet parameter values as security bounds	Rejected per Tan & Nahar 2026: a proof valid at a point value does not establish a robust domain. All device parameters are intervals with stated confidence, or UNCHARACTERIZED.
“Trényi & Curty NJP 2021” as correlation-treatment citation	Misattribution (D3 Finding F-D3-2): the real NJP 23, 093005 (2021) is a COW-QKD zero-error-attack paper. Substitute Yoshino 2018 / Zapatero 2021 / Sixto 2022 / Pereira 2025.

End of Deliverable 4. Companion: QO-D3-001 (Literature and Proof Review) supplies the verified citation records and assumption inventory on which every row above rests.